RIESZ BASES OF EXPONENTIAL FUNCTIONS IN L^2

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In this note we investigate conditions under which the system of functions $\{\exp i\lambda_\kappa t\}$, $-\infty<\kappa<\infty$, forms a Riesz basis in the space $L^2(-\sigma,\sigma)$. By $\{\lambda_\kappa\}$ we denote a noncondensing relatively dense set of points of the complex plane (these concepts are defined, e.g., in [1]) lying in the strip $|\operatorname{Im} z|\le\tau<\infty$, and by h we denote a real number such that $|h|>\tau$. The sequence $\{\lambda_\kappa\}$ is numbered in order of increasing real parts, and in the case when the real parts of several points coincide the numbering is arbitrary.

We partition the set $\{\lambda_\kappa\}$ by vertical lines into disjoint groups Λ_j, $-\infty<j<\infty$. The distance between the two adjacent lines that bound the group Λ_j is denoted by ℓ_j, and we require that the sequence $\{\ell_j\}$ be bounded. Such a construction will be called a Λ-partition of the set $\{\lambda_\kappa\}$. By K_j we denote the set of those κ for which $\lambda_\kappa\in\Lambda_j$. The symbols $\asymp$ and $\prec$ will be used in the same sense as in [1]. Supplementing the arguments of V. E. Katsnel'son used in [1], we obtain a generalization of the theorem proved there.

THEOREM 1. Suppose that 1) $\{\lambda_\kappa\}$ is the set of roots of an entire function of sine type with indicator diagram of width 2σ, and 2) $\{\delta_\kappa\}$ is any bounded sequence of complex numbers satisfying for some Λ-partition of the set $\{\lambda_\kappa\}$ the condition $|\sum_{\kappa\in K_j}\operatorname{Re}\delta_\kappa|\le d\ell_j$, $-\infty<j<\infty$, $d<1/4$. If the sequence $\{\lambda_\kappa+\delta_\kappa\}$ is separated, then the system $\{\exp i(\lambda_\kappa+\delta_\kappa)t\}$ forms a Riesz basis in $L^2(-\sigma,\sigma)$.

Following B. Ya. Levin, we define a function of the class C to be an entire function Φ of exponential type such that $(1+x^2)^{-1}\ln_+|\Phi(x)|\in L_1$. The following lemma describes the behavior of functions of the class C on lines parallel to the real axis.

LEMMA. Let $\{\delta_\kappa\}$ be a bounded sequence of complex numbers, and let $\{\lambda_\kappa\}$ and $\{\lambda_\kappa+\delta_\kappa\}$ be the sets of roots of entire functions of the class C (Φ and ψ, respectively). Then

$$\left|\frac{\psi(x+ih)}{\Phi(x+ih)}\right|\asymp\exp\operatorname{Re}\left\{-\sum_{|\lambda_\kappa|\le 2|x|}\left(\frac{\delta_\kappa}{\lambda_\kappa}+\frac{\delta_\kappa}{x+ih-\lambda_\kappa}\right)\right\}. \tag{1}$$

In the case $\lambda_\kappa=c\kappa$ this lemma was proved by B. Ya. Levin [2, p. 584].

Using Theorem 1 and the lemma, we can give some sufficient conditions for the system $\{\exp i\lambda_\kappa t\}$ to be a basis in terms of the behavior of an entire function with roots $\{\lambda_\kappa\}$. An example of such conditions is given in

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THEOREM 2. Suppose that 1) $\{\lambda_\kappa\}$ is the set of roots of an entire function Φ of exponential type with indicator diagram of width 2σ, 2) φ is a positive function on the semiaxis $[0,\infty)$ such that for $\gamma(x)=x[\varphi'(t)/\varphi(t)]_{t=|x|}$ the following conditions hold: $|\gamma(x)|\le\omega<1/2$, $\gamma'(x)=O(|x|^{-1})$. If $|\Phi(x+ih)|\asymp\varphi(|x|)$ and the sequence $\{\lambda_\kappa\}$ is separated, then the system $\{\exp i\lambda_\kappa t\}$ forms a Riesz basis in $L^2(-\sigma,\sigma)$.

A B-function is defined to be an entire function of exponential type with set of roots $\{\lambda_\kappa\}$ such that the system $\{\exp i\lambda_\kappa t\}$ forms a Riesz basis. On lines parallel to the real axis B-functions can behave in various ways. The condition 2) of Theorem 2 is satisfied, e.g., by such functions as x^α, $x^\alpha(\ln x)^\beta$, $(|\alpha|\le 1/2, |\beta|<\infty)$. The following example shows how strongly the modulus of a B-function can oscillate on the lines $\operatorname{Im} z=\text{const}$.

Let $\Psi(z)=z(\text{v.p.}\prod_{\kappa\ne0}[1-z(\kappa+\delta_\kappa)^{-1}]$, $|\delta_\kappa|\le d<1/4$. We set $\delta_\kappa=0$ for $\kappa\le0$, and $\delta_\kappa\ge0$, for $\kappa>0$. Then, applying the formula (1), we get:

$$|\Psi(x+ih)|\asymp\exp\operatorname{Re}\{-\sum_{1\le\kappa\le2|x|}\delta_\kappa[\kappa^{-1}+(x+ih-\kappa)^{-1}]\}.$$

The sum $\sum_{1\le\kappa\le2|x|}\delta_\kappa(x+ih-\kappa)^{-1}$ is bounded for $x<0$, and here $|\Psi(x+ih)|<1$. For natural numbers n

$$|\Psi(n+ih)|\asymp\exp\{-\sum_{\kappa=1}^{n}\kappa^{-1}(\delta_\kappa+\delta_{n-\kappa}-\delta_{n+\kappa})\}.$$

We now choose numbers $n_1, n_2, n_3, \ldots$, from the conditions $\sum_{\kappa=n_0}^{n_{p+1}}\kappa^{-1}>p+\sum_{\kappa=1}^{n_p}\kappa^{-1}$, $p=1,2,3,\ldots$, and we set $\delta_\kappa=0$ for $n_{2p-1}\le\kappa<n_{2p}$ and $\delta_\kappa=d$ for $n_{2p}\le\kappa<n_{2p+1}$. It is not hard to verify that in this case

$$|\Psi(n_{2p}+ih)|>\exp(2p-1)d,\quad |\Psi(n_{2p+1}+ih)|<\exp(-2pd).$$

Thus, on the semiaxis $[0,\infty)$ $|\Psi(x+ih)|$ oscillates with increasing amplitude.

Using Theorems 1 and 2, we can also get various results on the stability of the basis $\{\exp i\lambda_\kappa t\}$ under small parturbations. An example of assertions of this type is given by

THEOREM 3. Suppose that 1) Φ and φ are the functions described in the hypotheses of Theorem 2 and $|\Phi(x+ih)|\asymp\varphi(|x|)$, 2) $\{\delta_\kappa\}$ is a bounded sequence of complex numbers satisfying for some Λ-partition of the set $\{\lambda_\kappa\}$ the condition $|\sum_{\kappa\in\Lambda_j}\operatorname{Re}\delta_\kappa|\le b(1/2-a)$, $-\infty<j<\infty$, $b<1/2$. If the sequence $\{\lambda_\kappa+\delta_\kappa\}$ is separated, then the system $\{\exp i(\lambda_\kappa+\delta_\kappa)t\}$ forms a Riesz basis in $L^2(-\sigma,\sigma)$.

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LITERATURE CITED

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